ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 18	Lines of symmetry = $360 \div$ angle.
2	**(d)** 300	Multiply.
3	**(b)** A square	Only the square among these has 4.
4	**(d)** 60	The square fits after every multiple of 90°.
5	**(b)** 10	Add.
6	**(a)** 5	Count the letters with a fold line.
7	**(c)** 45	$360 \div$ spokes.
8	**(b)** 20	Lines of symmetry = $360 \div$ angle.
9	**(d)** 225	Multiply.
10	**(d)** hexagon	6 lines mean 6 sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sides $= 360 \div 40 = 9$ (1); Order $= 9$ (1).	1+1
12	square: 4 (1); regular octagon: 8; total 12 (1).	1+1
13	Reflect each x-coordinate: $x \to -x$ (1); New x values: -1, -3, -2 (1).	1+1
14	Sides $= 360 \div 36 = 10$ (1); Order $= 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 5 = 72^\circ$ (1); After 3 turns: 216° (1); Turns for full rotation $= 5$ (1).	1+1+1
16	H: order 2, lines 2; A: order 1, lines 1; T: order 1, lines 1 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 12 = 30^\circ$ (1); Lines of symmetry $= 12$ for a regular arrangement (1); Turn to mark 5: 150° (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); After 4 turns: 240° (1); Turns for full rotation $= 6$ (1).	1+1+1
18	O: order 2, lines 2; X: order 2, lines 2; Z: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 8 = 45^\circ$ (1); Lines of symmetry $= 8$ for a regular arrangement (1); Turn to mark 3: 135° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 5 (1); (b) 72° (1); (c) 5 (1); (d) 144° is 2 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: square: 90° (1); regular octagon: 45° (1); Difference $= 45^\circ$ (1); Yes, the larger count is a multiple of the smaller (2).	1+1+1+2
20	(a) 6 (1); (b) 60° (1); (c) 6 (1); (d) 180° is 3 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: equilateral triangle: 120° (1); regular hexagon: 60° (1); Difference $= 60^\circ$ (1); Yes, the larger count is a multiple of the smaller (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	The square fits after every multiple of $90°$.
2	**(d)** 8	Add.
3	**(b)** 2	Count the letters with a fold line.
4	**(c)** 40	$360 \div$ spokes.
5	**(a)** 24	Lines of symmetry = $360 \div$ angle.
6	**(d)** 216	Multiply.
7	**(a)** equilateral triangle	Equilateral triangle.
8	**(a)** 75	The square fits after every multiple of $90°$.
9	**(d)** 12	Add.
10	**(a)** 3	Count the letters with a fold line.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	equilateral triangle: 3 (1); regular hexagon: 6; total 9 (1).	1+1
12	Reflect each x-coordinate: $x \to -x$ (1); New x values: -2, -4, -3 (1).	1+1
13	Sides $= 360 \div 30 = 12$ (1); Order $= 12$ (1).	1+1
14	regular pentagon: 5 (1); square: 4; total 9 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 8 = 45°$ (1); After 5 turns: $225°$ (1); Turns for full rotation $= 8$ (1).	1+1+1
16	T: order 1, lines 1; I: order 2, lines 2; N: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 10 = 36°$ (1); Lines of symmetry $= 10$ for a regular arrangement (1); Turn to mark 4: $144°$ (1).	1+1+1
	OR: Angle $= 360 \div 9 = 40°$ (1); After 4 turns: $160°$ (1); Turns for full rotation $= 9$ (1).	1+1+1
18	M: order 1, lines 1; O: order 2, lines 2; S: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 6 = 60°$ (1); Lines of symmetry $= 6$ for a regular arrangement (1); Turn to mark 5: $300°$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 8 (1); (b) $45°$ (1); (c) 8 (1); (d) $135°$ is 3 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular pentagon: $72°$ (1); regular decagon: $36°$ (1); Difference $= 36°$ (1); Yes, the larger count is a multiple of the smaller (2).	1+1+1+2
20	(a) 9 (1); (b) $40°$ (1); (c) 9 (1); (d) $160°$ is 4 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: square: $90°$ (1); regular hexagon: $60°$ (1); Difference $= 30°$ (1); No, the larger count is not a multiple of the smaller (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 36	$360 \div$ spokes.
2	**(d)** 30	Lines of symmetry = $360 \div$ angle.
3	**(c)** 160	Multiply.
4	**(d)** isosceles triangle	Only turn of 360 degrees.
5	**(d)** 20	The square fits after every multiple of $90°$.
6	**(b)** 11	Add.
7	**(a)** 1	Count the letters with a fold line.
8	**(b)** 30	$360 \div$ spokes.
9	**(c)** 5	Count the letters with a fold line.
10	**(d)** 24	$360 \div$ spokes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflect each x-coordinate: $x \to -x$ (1); New x values: -3, -5, -4 (1).	1+1
12	Sides $= 360 \div 24 = 15$ (1); Order $= 15$ (1).	1+1
13	regular decagon: 10 (1); regular hexagon: 6; total 16 (1).	1+1
14	Reflect each x-coordinate: $x \to -x$ (1); New x values: -4, -6, -5 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 10 = 36°$ (1); After 7 turns: $252°$ (1); Turns for full rotation $= 10$ (1).	1+1+1
16	X: order 2, lines 2; Y: order 1, lines 1; Z: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 9 = 40°$ (1); Lines of symmetry $= 9$ for a regular arrangement (1); Turn to mark 2: $80°$ (1).	1+1+1
	OR: Angle $= 360 \div 12 = 30°$ (1); After 5 turns: $150°$ (1); Turns for full rotation $= 12$ (1).	1+1+1
18	S: order 2, lines 0; H: order 2, lines 2; O: order 2, lines 2 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 15 = 24°$ (1); Lines of symmetry $= 15$ for a regular arrangement (1); Turn to mark 4: $96°$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 10 (1); (b) $36°$ (1); (c) 10 (1); (d) $252°$ is 7 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular hexagon: $60°$ (1); regular 12-gon: $30°$ (1); Difference $= 30°$ (1); Yes, the larger count is a multiple of the smaller (2).	1+1+1+2
20	(a) 12 (1); (b) $30°$ (1); (c) 12 (1); (d) $150°$ is 5 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular pentagon: $72°$ (1); regular octagon: $45°$ (1); Difference $= 27°$ (1); No, the larger count is not a multiple of the smaller (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.